Buffers and default-low interface

Equal net names join. Exact connections and protection: assembly_connections sheets / READY_TO_WIRE.md. Ratings remain HOLD.

E2 DESIGN HOLD | Physical connections unapproved | Generic ports require actual-part mapping | 08 Nov 2026

U1 | SN74AHCT244N 3.3-to-5 V buffer

ADDITIONAL / MISSING OR SELECT

SN74AHCT244N

1 / 1OE_N	GND
2 / 1A1	L_FWD_3V3
3 / 2Y4	TR_5V
4 / 1A2	L_REV_3V3
5 / 2Y3	TL_5V
6 / 1A3	R_FWD_3V3
7 / 2Y2	TF_5V
8 / 1A4	R_REV_3V3
9 / 2Y1	ENABLE_5V
10 / GND	GND
11 / 2A1	ENABLE_3V3
12 / 1Y4	R_REV_5V
13 / 2A2	TF_3V3
14 / 1Y3	R_FWD_5V
15 / 2A3	TL_3V3
16 / 1Y2	L_REV_5V
17 / 2A4	TR_3V3
18 / 1Y1	L_FWD_5V
19 / 2OE_N	GND
20 / VCC	V5_LOGIC

R18 | DRV_A.ENB local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R32 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	ER_SAFE
2	GND

U2 | SN74LVC3G34DCU ECHO isolation

ADDITIONAL / MISSING OR SELECT

SN74LVC3G34DCU

1 / 1A	EF_DIV
2 / 3Y	ER_SAFE
3 / 2A	EL_DIV
4 / GND	GND
5 / 2Y	EL_SAFE
6 / 3A	ER_DIV
7 / 1Y	EF_SAFE
8 / VCC	V3V3

R14 | DRV_L.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R16 | DRV_R.L_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R30 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EF_SAFE
2	GND

R5 | ENABLE input pull-down

ADDITIONAL / MISSING OR SELECT

2.2k 1%, >=1/8 W

1	ENABLE_3V3
2	GND

R13 | DRV_L.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R15 | DRV_R.R_EN local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R17 | DRV_A.ENA local EN pull-down

ADDITIONAL / MISSING OR SELECT

10k 1%, >=1/8 W

1	EN_DRIVER_5V
2	GND

R31 | ECHO output default-low

ADDITIONAL / MISSING OR SELECT

100k 1%, >=1/8 W

1	EL_SAFE
2	GND

Red +supply | Black GND | Blue control | Green sensor | Orange motor | Thick = high-current | Dashed = missing/blocked

Net labels join across sheets. Crossings alone do not join. No component footprint / connector orientation is implied.